

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 8 Winter 2025 (Supply.)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 8

Subject Code & Name: BTCVSS801B - GEOSYNTHETICS AND REINFORCED SOIL STRUCTURES

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12
- 1 A nonwoven geotextile is preferred over a woven geotextile for filtration because it:
A) Has higher tensile strength
B) Provides higher apparent opening size (AOS)
C) Exhibits greater thickness and three-dimensional pore structure
D) Is chemically more resistant
 - 2 In pull-out tests, increasing reinforcement length beyond a certain limit results in:
A) Linear increase in pull-out resistance
B) Reduction in mobilized friction
C) No significant increase in pull-out capacity
D) Sudden tensile rupture
 - 3 According to Indian standards, which IS code provides design guidelines for reinforced soil retaining structures?
A) IS 456
B) IS 800
C) IS 14458
D) IS 1893
 - 4 During external stability analysis, reinforced soil mass is treated as:
A) A rigid gravity wall
B) A flexible retaining structure
C) A composite homogeneous block
D) A drained soil mass only
 - 5 Reinforcement spacing in reinforced soil walls is primarily controlled by:
A) Facing type
B) Tensile strength of reinforcement
C) Allowable tensile force and earth pressure distribution
D) Wall height only

- 6 Infinite slope analysis is unsuitable for reinforced soil slopes when:
 - A) Slope angle is small
 - B) Failure surface intersects reinforcement layers
 - C) Soil is cohesionless
 - D) Water table is absent
- 7 Bilinear wedge analysis is particularly useful when:
 - A) Soil is homogeneous
 - B) Failure surface changes inclination within reinforced zone
 - C) Reinforcement spacing is uniform
 - D) Backfill is horizontal
- 8 Reinforced soil beneath footings behaves as:
 - A) Elastic foundation
 - B) Rigid slab
 - C) Composite material with improved stress distribution
 - D) Purely frictional soil
- 9 Geosynthetic encasement in stone columns improves load capacity by:
 - A) Increasing drainage path
 - B) Preventing lateral bulging
 - C) Increasing stone density
 - D) Reducing column diameter
- 10 Clogging potential of geotextile filters increases when:
 - A) Soil is well graded
 - B) Geotextile permeability is very high
 - C) Soil contains high fines content
 - D) Flow is laminar
- 11 Erosion control mats are most effective when:
 - A) Installed after vegetation growth
 - B) Used without anchorage
 - C) Combined with vegetation
 - D) Placed under water
- 12 Stress cracking in geomembranes is primarily associated with:
 - A) Short-term loading
 - B) High permeability soils
 - C) Sustained tensile stress and chemical exposure
 - D) UV radiation only

Q2. Solve the following.

- A) Define geosynthetics. Briefly explain the different types of geosynthetics and their applications. <https://www.batuonline.com> 6
- B) What are the different tests conducted on geosynthetics? Briefly explain any two. 6

Q3. Solve the following.

- A) Explain the different types of soil retaining structures and highlight the advantages of geosynthetic reinforced soil retaining walls. 6
- B) Explain the concept of seismic loading in reinforced soil retaining walls. 6

Q4. Solve Any Two of the following.

- A) Explain the step-by-step design procedure of a reinforced soil retaining wall with simple geometry. 6
- B) Explain the role of drainage in the design and performance of reinforced soil retaining walls. 6

- C) What are the design parameters and procedures for reinforced soil supporting shallow foundations? 6
- Q5. Solve Any Two of the following.
- A) Explain the concept of bilinear wedge analysis for reinforced soil slopes and its significance. 6
- B) What are Geosynthetic Encased Stone Columns (GESCs)? What are their benefits for load support in soft soils? 6
- C) What are the filtration criteria for geosynthetics used in soil filtration applications? 6
- Q6. Solve Any Two of the following.
- A) How do geosynthetics contribute to erosion control? Mention the types commonly used. 6
- B) What are natural geosynthetics? Discuss their applications and limitations. 6
- C) What is a landfill liner system and how are geosynthetics used in it? 6

*** End ***